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28 February 1984 and
AMENDMENT 1
26 July 1984

MILITARY SPECIFICATION

# CARTRIDGE, IMPULSE, CCU-22/A/A ASSEMBLY

*This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.*

## 1. SCOPE

**1.1 Scope.** This specification covers the minimum requirements for the procurement of the CCU-22/A/A impulse cartridge assembly referred to herein as the cartridge assembly.

## 2. APPLICABLE DOCUMENTS

### 2.1 Government documents.

**2.1.1 Specifications and standards.** The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-P-116 | Preservation, Methods of |
| MIL-S-9479 | Seat System, Upward Ejection, Aircraft, General Specification for |
| MIL-P-82652 | Propellant, Solid, Composite |
| MIL-C-82734 | Catapult, Aircraft Ejection Seat, CKU-5/A/A Assembly |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, Maryland 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC NO. N4455 FSC 1377

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STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking for Shipping and Storage |
| MIL-STD-414 | Sampling Procedures for Inspection by Variables |
| MIL-STD-453 | Inspection, Radiographic |
| MIL-STD-810 | Environmental Test Methods |
| MIL-STD-1167 | Ammunition Data Cards |
| MIL-STD-1168 | Ammunition Lot Numbering |
| DOD-STD-2101 | Classification of Characteristics |

**2.1.2 Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

CODE OF FEDERAL REGULATIONS

| | |
|---|---|
| 49 CFR 100-199 | Transportation |

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 53711)

| | |
|---|---|
| 5184323 | Cartridge, Impulse, CCU-22/A/A Assembly |
| 5184324 | Shear Pin |

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

**2.2 Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 2200 Mill Road, Alexandria, VA 22314.)



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UNIFORM CLASSIFICATION COMMITTEE, AGENT
(Uniform Freight Classification)

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606).

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

**2.3 Order of precedence.** In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

**3.1 First article.** Unless otherwise specified in the contract or purchase order (see 6.2), first article samples of the cartridge assemblies are required and shall be manufactured using the methods and procedures proposed for production. The samples shall be tested as specified in section 4 for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing cartridge assemblies that comply with the technical requirement of the contract (see 4.4).

**3.1.1 Propellant aging.** Three blocks of cured propellant from each batch of propellant used in the first article sample shall be submitted to the Naval Ordnance Station, Indian Head, Maryland 20640-5000, Attn: Code 5120, for aging testing. High temperature aging shall be conducted for 84 days with an atmosphere of dry air and at 165 + 5 °F. At zero time and at the end of the 28-, 56-, and 84-day aging, physical properties shall conform to MIL-P-82652. The rate for stress/strain determination shall be 2.0 inches/minute. The sample shall be cast into rectangular cardboard containers approximately 1.5 inches high by 8 inches wide by 11 inches long. All samples submitted shall be bare propellant, individually wrapped in moisture-proof packaging, and properly identified as to propellant lot and batch number.

**3.2 Materials.** All materials, parts, and assemblies shall be in accordance with Drawings 5184323, 5184324, this specification, and the documents referenced herein.

**3.2.1 Primary components.** Only primary components from a single primary component lot shall be used in a production lot of cartridge assemblies.

**3.3 Age.**

**3.3.1 Explosive components.** All primers shall have been manufactured and certified within 24 months or recertified within 12 months of the assembly date for the cartridge assembly lot. All other explosive components, including but not limited to igniters, shall have been manufactured or recertified within 12 months of the assembly date.

**3.3.2 Propellant.** The propellant shall be manufactured within the 150 days prior to the assembly date for the cartridge assembly lot. When manufactured as a subassembly of the CKU-5/A/A Rocket Catapult in accordance with MIL-C-82734, the propellant shall be manufactured within the 12 months prior to the assembly date for the rocket catapult assembly lot.

**3.4 Ballistic performance requirements.** The cartridge assemblies shall meet the ballistic performance requirements of table I when assembled into acceptable rocket catapult hardware in accordance with MIL-C-82734 and test fired using an ejected weight of 375 pounds.



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TABLE I. Ballistic performance requirements (375 pound ejected weight).

| Parameter | -65 °F | | 70 °F | | 165 °F | | AQL |
|---|---|---|---|---|---|---|---|
| | MIN | MAX | MIN | MAX | MIN | MAX | |
| Dynamic Response Index (DRI)[1] | — | 18.0 | — | 18.0 | — | 22.0 | 0.65 |
| Separation velocity (ft/s)[2] | 35 | — | 35 | — | 35 | — | 1.0 |
| Ignition delay (ms) | — | 30 | — | 30 | — | 30 | 0.65 |
| Axial thrust (lbf) | — | 7000 | — | 7000 | — | 7000 | 1.0 |
| Pressure (psi) | 1200 | — | 1200 | — | 1200 | — | 1.0 |

[1]DRI is defined in MIL-S-9479 (see 6.4.3)
[2]Nominal separation velocities at 70 and 165 °F are 42.0 ft/s and 47.0 ft/s respectively.

**3.5 Propellant.** The propellant shall be in accordance with MIL-P-82652.

**3.6 Radiographic requirements.** Each loaded cartridge assembly shall be examined radiographically to ascertain the presence, condition, and correct placement of internal components.

**3.7 Shear pin strength.** Each lot of shear pins in accordance with Drawing 5184324 shall be tested in accordance with 4.5.7. Shear pins shall shear (double shear) when subjected to a load of 36-41 pounds applied at a rate of 0.125 inches per minute.

**3.8 Protective finish.** Protective finish shall be in accordance with the applicable specification and drawings and shall apply to all surfaces of the component regardless of contour configuration.

**3.9 Environmental requirements.** The cartridge assemblies shall withstand the environmental conditioning of 3.9.1 through 3.9.3. Cartridges subjected to this conditioning shall show no evidence of deterioration of propellant or disarrangement of components when examined in accordance with 4.5.6 and shall meet the ballistic performance requirement of 3.4 when test fired in accordance with 4.4.2.

**3.9.1 High temperature.** The cartridge assembly shall withstand the high temperature test of 4.4.1.1.

**3.9.2 Vibration.** The cartridge assembly shall withstand the vibration test of 4.4.1.2.

**3.9.3 Temperature shock.** The cartridge assembly shall withstand the temperature shock test of 4.4.1.3.

**3.10 Identification marking.** Each cartridge assembly shall be marked in accordance with Drawing 5184323.

**3.10.1 Production lot designation.** The production lot number shall follow MIL-STD-1168. The construction of the lot number shall be as follows:

XXX75D002-007
a b c d e

where

a = Manufacturer's Identification Code
b = The year of manufacture of the oldest batch of propellant in the lot
c = The month of manufacture of the oldest batch of propellant in the lot expressed as an alpha code in accordance with 4.1.3 of MIL-STD-1168
d = Lot interfix number (controlled by Naval Ordnance Station, Code 5120, Indian Head, MD 20640-5000 or procuring activity)
e = Lot sequence number - lot number (manufacturer is responsible for the assignment of lot numbers).



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**3.11 Workmanship.** The workmanship shall be that required by the applicable drawings, referenced specifications, and the best industrial practices governing the quality production of interchangeable parts for maintaining the dimensions, finishes, tolerances, and quality specified herein. All components and assemblies shall be free from burrs, contamination, corrosion, sharp edges, or foreign material which could result in malfunction of the unit or components or be a safety hazard in handling.

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Responsiblity for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

**4.1.1 Responsiblity for compliance.** All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

**4.2 Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

**4.3 Inspection conditions.** Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 65 to 95 °F
b. Altitude: Normal ground
c. Vibration: None
d. Humidity: Room ambient to 95 percent relative, maximum.

**4.4 First article inspection.** Unless otherwise specified in the contract (see 6.2), a first article sample of 21 cartridge assemblies, manufactured in accordance with 3.1, shall be submitted to the Naval Ordnance Station, Indian Head, MD 20640-5000 for first article inspection. The first article samples shall be subjected to the quality conformance inspection of 4.5, with the exception of the test firing of 4.5.5, followed by environmental testing, radiographic examinations, and static firing in the sequence specified in table II. Acceptance of the first article sample shall be based on no defects in the sample. Any production of the cartridge assemblies by the contractor, prior to approval of the first article sample, shall be at the contractor's risk. First article samples shall not be included as a part of the quantity specified for delivery in the contract (see 6.2).

**4.4.1 Environmental tests.**

**4.4.1.1 High temperature.** The cartridge assembly shall be subjected to the high temperature storage test in accordance with Method 501 of MIL-STD-810. The test items shall be maintained at 165 °F for 50 hours minimum. Operation of the cartridge assembly is not required.



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TABLE II. First article evaluation test program.[1]

| Test Para. | Test | Unit number | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 |
| 4.5.6 | Radiographic inspection | a | a | a | a | a | a | a | a | a | a | a | a | a | a | a | a | a | a | a | a | a |
| 4.4.1.1 | High temperature | b | b | b | b | b | b | b | b | b | b | | | | | | | | | | | |
| 4.4.1.2 | Vibration −65 °F | | | | | | | | | | | b | b | b | b | b | b | | | | | |
| 4.4.1.2 | Vibration 165 °F | | | | | | | | | | | | | | | | | b | b | b | b | b |
| 4.5.6 | Radiographic inspection | c | c | c | c | c | c | c | c | c | c | c | c | c | c | c | c | c | c | c | c | c |
| 4.4.1.3 | Temperature shock | d | d | d | d | d | d | d | d | d | d | d | d | d | d | d | d | d | d | d | d | d |
| 4.5.6 | Radiographic inspection | e | e | e | e | e | e | e | e | e | e | e | e | e | e | e | e | e | e | e | e | e |
| 4.4.2 | Static firing −65 °F | f | f | f | f | | | | | | | f | f | | | f | | | f | f | | |
| 4.4.2 | Static firing 70 °F | | | | | f | f | | | | | | | | | | f | | | | | |
| 4.4.2 | Static firing 165 °F | | | | | | | f | f | f | f | | | f | f | | | f | | | f | f |

[1]Letters denote sequence of testing.

4.4.1.2 **Vibration.** The cartridge assembly shall be subjected to random vibration, for jet aircraft, in accordance with Method 514 of MIL-STD-810 for 6 hours with 2 hours on each axis. Before vibration, units shall be temperature conditioned in accordance with table II. The input vibratory acceleration shall have a Wo factor of 0.1 $g^2$/Hz. If the cartridge assembly must be removed from the environmental conditioning chamber or if the temperature falls outside the specified temperature limits for a period of more than 5 minutes, the testing shall be stopped, and the unit shall be reconditioned. A reconditioning time of 2 minutes for every minute the unit is outside the specified temperature envelope shall be required. A continuous temperature history of the unit shall be kept for all conditioning and testing time. Unless otherwise noted, the cartridge assembly shall be mounted on a test fixture in a manner that will simulate actual seat/aircraft installation and excited through the attachment fittings.

4.4.1.3 **Temperature shock.** The cartridge assembly shall be subjected to the temperature shock test in accordance with Method 503 of MIL-STD-810. The test temperature extremes shall be −65 and 165 °F, and the duration of exposure at each temperature shall be 8 hours. Operation of the cartridge assembly is not required.

4.4.2 **Static firing test.** Within 30 days after completion of environmental exposure, the cartridge assemblies shall be static fired at the Naval Ordnance Station, Indian Head, MD 20640-5000. The sample shall be temperature conditioned per table II for a minimum of 8 hours, and test fired within 10 minutes after removal from the conditioning chamber to verify conformance to the requirements of 3.4. Data specified in table I shall be analyzed in accordance with MIL-STD-414, Section B Part 1, using the procedures of B-5 through B-7. Nonconformance with table I shall be cause for rejection of the first article sample.

4.4.3 **Propellant aging.** Propellant samples shall be prepared and tested in accordance with the requirements of 3.1.1. Test shall be performed in accordance with MIL-P-82652. Failure of the propellant to meet the requirements of MIL-P-82652 shall be cause for rejection of the batch.

4.4.4 **First article propellant morphology.** Four cartridge case and grain assemblies from each batch of propellant used in the first article sample are to be submitted to the Naval Ordnance Station, Indian Head, Maryland 20640-5000, Attn: Code 5120, for morphological examination of the propellant. This examination shall include microscopy and chromatography. All samples shall be individually wrapped in moisture-proof packaging and properly identified as to propellant lot, batch number, and grain serial number. These tests will be used to determine propellant homogeneity and will be used as baseline information.



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**4.5 Quality conformance inspection.** The quality conformance inspection shall consist of the examinations and tests of 4.5.4 through 4.5.7 conducted on samples selected as specified in 4.5.2. The shear pin test of 4.5.7 shall be performed prior to installation of any shear pins from the lot in question into cartridge assemblies. The contractor is responsible for assuring that all cartridge assemblies offered for acceptance meet all the requirements of the applicable drawings and specifications. Items of inspection shall be serialized and inspection data traceable to a specific serial number.

**4.5.1 Test reports.** When specified in the contract or order (see 6.2.2), the contractor shall furnish test reports to the procuring activity. Test reports shall include the following information:

a. Propellant test results (see 3.5)
b. Static firing performance test results (see 4.5.5)
c. Summary sheet
d. Ammunition data cards
e. Propellant/assembly production records
f. Certification of age of explosive components and propellant (see 3.3).

**4.5.1.1 Propellant test reports.** The following propellant information shall be submitted with each production lot. The materials and components of each propellant batch shall be listed with the manufacturer, lot number, and date of manufacture. The burning rates at 2500, 3500, 4500, 5000, and 7500 psi performed at −65, 70, and 165 °F for each pressure (three samples at each temperature and pressure per mix) shall also be included. In addition, include the results of tensile strength and elongation tests at −65, 70, and 165 °F with a pull rate of 2 inches/minute (three samples at each temperature per mix), heat of explosion (three samples per mix), and verification that ingredients were properly weighed and added to each mix (mix sheet(s)).

**4.5.1.2 Static firing performance test reports.** All computations and data required for ballistics evaluation, generated during quality conformance inspection static firings per 4.5.5, shall be reported, including firing traces. The following information shall also be listed:

a. Grain designation, manufacture, lot number, and date of propellant manufacture
b. Type of test equipment
c. Deviations from standard testing procedures
d. Additional comments (malfunctions, unusual occurrences)
e. Disassembly findings report (see 4.5.5.2).

**4.5.1.3 Summary sheet.** A summary of the results of the tests for the parameters listed in table I shall be submitted to the designated Government activity for each inspection lot. The summary shall include the sample size, number and nature of defects found, and the disposition (accepted or rejected).

**4.5.1.4 Ammunition data cards.** As specified in the contract, ammunition data cards (DD Form 1650) shall be completed in accordance with MIL-STD-1167 for each cartridge assembly lot. Copies shall be provided in accordance with contract requirements and one copy shall be forwarded to the Commanding Officer, Naval Ordnance Station, Indian Head, MD 20640-5000 (Code 5120).

**4.5.1.5 Propellant/assembly production records.** The contractor shall prepare and retain for the service life of the unit propellant production records including batch number, mix designation, rocket motor lot number, rocket motor serial number, cartridge lot number, cartridge case and grain assembly serial number, and cartridge serial number.



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**4.5.2 Sampling for quality conformance inspection.**

**4.5.2.1 Lot.** For the purposes of identification, inspection, and shipment, a lot of loaded assemblies shall consist of 420 assemblies, maximum, inclusive of test samples. The homogeneity of the inspection lot shall be retained by consisting of cartridge assemblies that are manufactured at one time with one propellant lot and one primary component lot.

**4.5.2.2 Sampling for examination.** Designation of defects (critical, major, and minor) shall be in accordance with DOD-STD-2101. Samples for examination shall be selected in accordance with inspection level II of MIL-STD-105.

**4.5.2.3 Sampling for static firing test.** A sample of cartridge assemblies shall be submitted to the Naval Ordnance Station, Indian Head, MD 20640-5000 for static firing in accordance with 4.5.5. The sample size shall be in accordance with table III. Cartridge assembly samples shall not be preselected during production. Cartridge assemblies selected for samples shall not be included as a part of the quantity specified for delivery in the contract or order. Only primary components from a single primary component lot shall be used in a production lot of cartridge assemblies. One primary component lot may be used in more than one cartridge assembly production lot. The ballistic sample size shall be divided as equally as possible among cartridge batches. From each cartridge batch not represented in the sample cartridge assemblies, two sample cartridges shall be selected for static firing by the contractor (see 4.5.5.1), so that at least two cartridges from each cartridge batch have been static fired.

Table III. Sample size for firing test.

| Lot size | Total firing sample | Sample per temperature extreme |
|---|---|---|
| Up to 110 | 10 | 5 |
| 111-180 | 14 | 7 |
| 181-300 | 20 | 10 |
| 301-340 | 22 | 11 |
| 341-380 | 24 | 12 |
| 381-420 | 26 | 13 |

**4.5.2.4 Sampling for radiographic inspection.** Each cartridge assembly shall be subjected to the radiographic inspection specified in 4.5.6.

**4.5.2.5 Sampling for shear pin test.** A sample of 50 shear pins from each shear pin lot shall be subjected to the shear pin strength test of 4.5.7.

**4.5.3 Resubmission of rejected lot.** A sample of cartridge assemblies shall be selected in accordance with 4.5.2.3 and submitted when, with prior approval of the procuring activity, the contractor requests that the rejected lot be test fired under different conditions or the contractor believes a rejected lot may be reworked to meet the requirements of this specification.

**4.5.4 Examination.** Sample cartridge assemblies selected in accordance with 4.5.2.2 shall be examined for defects in accordance with the applicable drawings, 3.8, 3.10, and 3.11. The acceptable quality levels (AQL's) per MIL-STD-105 shall be as follows:

a. Critical defects: 0.04 percent defective
b. Major defects: 1.0 percent defective
c. Minor defects: 2.5 percent defective



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**4.5.5 Production lot acceptance test firing.** Sample cartridge assemblies selected in accordance with 4.5.2.3 shall be static fired at the Naval Ordnance Station, Indian Head, MD 20640-5000 within 30 days after submission for acceptance of the cartridge assembly lot. The samples shall be divided into two equal groups. One group shall be conditioned at −65 °F and one at 165 °F for a minimum of 8 hours and static fired within 10 minutes after removal from the conditioning chamber. Minimum equipment accuracies shall be as shown in table IV. Data specified in table I shall be analyzed in accordance with MIL-STD-414, Section B, Part 1, using the procedure of B-5 through B-7. Nonconformance with table I shall be cause for rejection of the lot.

NOTE

When manufactured by the same contractor as a subassembly of the CKU-5/A/A rocket catapult in accordance with MIL-C-82734, a separate lot acceptance test is not required for the cartridge assembly. However, separate batch acceptance of cartridge batches not represented in the rocket catapult assembly acceptance tests must be tested in accordance with 4.5.2.3.

Table IV. Test equipment accuracies.

| Requirement | Equipment accuracy |
|---|---|
| Maximum thrust | ±2.0 percent |
| Time | ±0.001 second |
| Maximum acceleration | ±5.0 percent |
| Pressure | ±2.0 percent |
| Temperature | ±2.0 °F |

**4.5.5.1 Contractor batch acceptance test firing.** Additional sample cartridges selected by the contractor to meet the requirement of 4.5.2.3 shall be static fired by the contactor as cartridge batch acceptance tests. An equal number of cartridges shall be fired at each temperature: −65 and 165 °F. Failure of any cartridge to meet the requirements of table I shall be cause for rejection of the specific batch of cartridges.

**4.5.5.2 Postfiring disassembly.** A postfiring disassembly and inspection shall be conducted on all first article, lot acceptance, and batch acceptance test units. Detailed observations on conditions, including marginal conditions, of individual components shall be noted. When specified in the contract (see 6.2), a disassembly findings report shall be submitted to the contracting activity.

**4.5.6 Radiographic inspection.** Cartridge assemblies shall be radiographically inspected in accordance with MIL-STD-453, radiographic quality level 1. Any deficiency shall be cause for rejection of the unit. Radiographs shall be capable of detecting defects and cartridge assembly components 0.03 inch in length or width, whichever is smaller. When specified in the contract (see 6.2), all radiographs taken of the cartridge assemblies or components thereof shall be forwarded to the activity designated by the contracting activity. Otherwise, all radiographs shall be inspected on-site by a designated Government representative. Acceptance of the cartridge assembly lot shall be subject to approval of the radiographs by the designated Government activity. All radiographs of accepted lots of cartridge assemblies are to be retained by the contractor for 8 years or 1 year past the service life of the unit, whichever is longer.

**4.5.7 Shear pin test.** A load shall be applied at a rate of 0.125 inch per minute to each sample shear pin selected in accordance with 4.5.2.5. If any shear pin fails to shear at 36-41 pounds, the entire lot of shear pins shall be rejected, and no pin from that lot shall be used in any catapult assembly.

**4.6 Packaging inspection.** The packaging, packing, and marking shall be inspected to verify conformance with the requirements of section 5.



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## 5. PACKAGING

**5.1 Preservation, packaging, and packing.** Unless otherwise specified in the contract (see 6.2), preservation, packaging, and packing shall be level A.

**5.1.1 Level A.** Preservation, packaging, and packing shall be in accordance with the applicable drawings and MIL-P-116, as specified in the contract (see 6.2). Containers shall conform to Uniform Freight Classification, National Motor Freight Classification, or to rules of other carriers applicable to the mode of transportation.

**5.2 Marking.** In addition to any special markings required by the contract (see 6.2), each container shall be marked in accordance with MIL-STD-129 and Code of Federal Regulations 49 CFR 171-179 (see 6.8.1).

## 6. NOTES

**6.1 Intended use.** The CCU-22/A/A impulse cartridge assembly is intended to be used with the CKU-5/A/A aircraft ejection seat catapult assembly in the F-15, F-16, A-10, and B-1 aircraft.

**6.2 Ordering data.**

**6.2.1 Acquisition requirements.** Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity required (exclusive of first article and static firing test samples) (see 4.4 and 4.5.2.3).

c. When a first article sample is not required (see 3.1, 4.4, and 6.5).

d. Assigned activity for first article inspection and static firing tests (see 4.4, 4.4.2, 4.5.2.3, 4.5.5, 4.5.6, and 6.7).

e. Applicable drawing number(s) and/or methods and procedures of MIL-P-116 to be used for level A packaging (see 5.1.1).

f. Packaging requirements if other than level A (see 5.1).

g. Any special markings required (see 5.2 and 6.8.1).

h. Safety precautions (see 6.8).

i. Disposition of hardware, scrap propellant, and metal parts (see 6.9).

j. Inspection conditions if other than as specified (see 4.3).

**6.2.2 Data requirements.** When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

| Paragraph No. | Data requirement title | Applicable DID No. |
|---|---|---|
| 4.5.1 | Reports, Test | DI-T-2072 |
| 4.5.1 | Ammunition Data Cards | DI-MISC-80043 |
| 4.5.1 | Certification Data/Report | UDI-A-23264B |



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(Data item descriptions related to this specification, and identified in this section will be approved and listed as such in DOD 5010.12-L, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

**6.3 Definitions.**

**6.3.1 Catapult ignition delay time.** Catapult ignition delay time is the elapsed time from the first indication of shearing of the firing pin shear pins to first indication of catapult thrust.

**6.3.2 Catapult separation velocity.** Catapult separation velocity is defined as the average velocity of the simulated ejected weight propelled horizontally at catapult separation measured over the interval separation to 12 inches after separation.

**6.3.3 Catapult dynamic response index (DRI).** DRI is representative of the maximum dynamic compression of the vertebral column of the human body, as defined in MIL-S-9479.

**6.3.4 Propellant batch.** A propellant batch is a mix of ingredients, each ingredient composed of one homogeneous lot, that is mixed at one time in one vessel, resulting in a homogeneous mixture.

**6.3.5 Propellant lot.** A propellant lot is a group of one or more propellant batches, all using the same lot of each ingredient.

**6.3.6 Catapult cartridge batch.** A catapult cartridge batch is all cartridges cast from one propellant batch.

**6.3.7 Catapult cartridge lot.** A catapult cartridge lot is a group of cartridges manufactured at one time and composed of one propellant lot and one lot of primary components.

**6.3.8 Primary components.** Primary components are all components containing explosive ingredients, including the cartridge percussion primers, $BKNO_3$ pellets, cartridge case and grain assemblies, propellant shavings, and shear pins.

**6.4 First article.** When a first article inspection is required, the CCU-22/A/A impulse cartridge assembly should be a first article sample. The first article should consist of 21 units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

**6.5 Pressure transducers.** Instructions for adaption of first article catapult assemblies for pressure transducers should be compatible with instrumentation that has a response of not less than 600 Hz and that is capable of detecting ignition peak pressures and unstable burning areas.

**6.6 Safety precautions.** The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M, are applicable and should be specified in the contract or order as required by the DOD FAR Supplement, Part 28, Subpart 28.7102.

NOTE

When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP 5, are applicable.



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**6.6.1 Explosive hazard classification.** The explosive hazard classification for this cartridge assembly has been designated as:

Explosive Hazard Classification — Class 1 Division 4
Storage Compatibility Group — Group C
Department of Transportation (DOT) Hazard Class — Class C
DOT Marking — Propellant Explosive

**6.6.2 Hazard notice.** The cartridge assembly described herein is flammable and/or explosive and consequently presents a hazard in manufacture, handling, storage, and shipment. The contractor should recognize this hazard and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidence.

**6.7 Disposition of hardware, scrap propellant, and metal parts.** The contract should specify that all items purchased become the property of the U.S. Government; no hardware, propellant, or metal parts acceptable, scrapped, or tested be disposed of without authorization of the contracting activity; all items should be made available to the procuring activity upon request; and that the disposal of any tested hardware, propellant, or metal parts without authorization by the procuring activity will result in the test being declared invalid, and a retest will be required.

**6.8 Life.** The cartridge assembly has a design life of 7 years with an installed service life of 7 years from the date of propellant manufacture. Design life includes both the installed service life and the shelf life.

**6.9 Changes from previous issue.** Asterisks are not used in this revision to identify changes with respect to previous issue due to the extensiveness of the changes.

**6.10 Subject terms (keyword) listing.**

Ejection equipment, aircraft
Propellant
Propellant actuated device
Safety equipment, aircrew

Custodians:
Navy-OS
Air Force-70

Preparing Activity:
Navy-OS
(Project 1377-N925)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-C-82733A | CARTRIDGE, IMPULSE, CCU-22/A/A Assembly |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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